

Arjun Giriraj Day 14

1. What does Nmap stand for?
Network Mapper.
2. What is Nmap used for?
Nmap is used to gather information about hosts, ports, services, network exposures.
3. What is network discovery?
To identify which hosts are active and responsive on the network.
4. What is a port scan?
To check the ports and their states with their services.
5. What does an open port generally indicate?
The open ports generally indicate that the port is accepting suitable connections.
6. Does an open port automatically mean vulnerability? Explain.
Open port does not mean vulnerability, Real risks depends on the service, its version, its configuration, authenticity and patch status.
7. What does closed mean conceptually?
Closed means the host is reachable but no service is accepting the connection.
8. What does filtered mean conceptually?
Filtered means nmap does not determine the state because of the network filtering like firewalls, security control, network policy.
9. What does nmap 127.0.0.1 do?
A basic scan on your own localhost. Scan shows ports, state, service.
10. What does -p do?
-p selects which ports to check.
11. What does -sV do?
To check the service and version information on target.
12. What does -oN do?
It saves normal readable output to a file.
13. Why should scan results be documented?
Because for a real security professional documentation is a good habit. It helps to keep record for the targets.
14. What is cybersecurity scope?
Scope is what we could assess. It can be "Inscope" our own vm, "Outscope" everything else.
15. Why must authorization exist before scanning?

Because scanning or finding vulnerability without any authorization is illegal.

16. What is the difference between scanning and exploitation?

Scanning is the first part to gather information about target; Exploitation is the part where the attacker or security professional takes advantage of vulnerability.

17. Why is localhost a good beginner target?

Because it is our own machine making it controlled environment which might help to learn nmap without scanning the unauthorized target.

18. How do ports relate to services?

Services listen for connection on specific ports.

19. Why are unexpected listening services worth reviewing?

Unexpected services may increase the system's attack surface that's why their configuration, version, authentication, and patch status worth reviewing.

20. Explain a responsible Nmap assessment workflow.

Nmap assessment workflow

Obtain authorization → define scope → identify targets → perform discovery → scan ports → identify services/versions → document findings → review exposure → recommend remediation.

PRACTICAL SCREENSHOTS (localhost only):

nmap --version · nmap 127.0.0.1 · nmap -p 22,80,443 127.0.0.1 · nmap -sV 127.0.0.1 ·
nmap -oN day14-scan.txt 127.0.0.1 · cat day14-scan.txt

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venom@VENOM:~$ nmap --version
Nmap version 7.99 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.8 openssl-3.6.3 libssh2-1.11.1 libz-1.3.1 libpcap-1.10.4
libpcap-1.10.6 nmap-libdnet-1.18.0 ipv6
Compiled without:
Available nsock engines: epoll poll select
venom@VENOM:~$ nmap 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-22 18:01 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000020s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.08 seconds
venom@VENOM:~$ nmap -p 22,80,443 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-22 18:01 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000030s latency).

PORT      STATE SERVICE
22/tcp    closed ssh
80/tcp    closed http
443/tcp    closed https

Nmap done: 1 IP address (1 host up) scanned in 0.08 seconds
```

```

venom@VENOM:~$ nmap -sV 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-22 18:04 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000020s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.19 seconds
venom@VENOM:~$ nmap -oN dayl4-scan.txt 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-22 18:05 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000020s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.07 seconds
venom@VENOM:~$ cat dayl
dayl0 lab/      dayl4-scan.txt
venom@VENOM:~$ cat dayl4-scan.txt
# Nmap 7.99 scan initiated Tue Sep 22 18:05:08 2026 as: /usr/lib/nmap/nmap --pri
vileged -oN dayl4-scan.txt 127.0.0.1
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000020s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

# Nmap done at Tue Sep 22 18:05:08 2026 -- 1 IP address (1 host up) scanned in 0
.07 seconds

```

Also include your MINI ASSESSMENT REPORT (lab Task 10): target, authorization, open ports, services, observations, recommendation.

1. TARGET: 127.0.0.1
2. AUTHORIZED: YES
3. OPEN PORTS: none
4. SERVICES: ssh, http, https
5. OBSERVATIONS: The localhost host was reachable, but ports 22, 80, and 443 were closed.